ENGINE TEMPERATURE GAUGE

DESCRIPTION

Fig. 12 Engine Coolant Temperature Icon

An engine coolant temperature gauge is standard equipment on all instrument clusters. The engine coolant temperature gauge is located in the lower right quadrant of the instrument cluster, below the oil pressure gauge. The engine coolant temperature gauge consists of a movable gauge needle or pointer controlled by the instrument cluster circuitry and a fixed 90 degree scale on the cluster overlay that reads left-to-right from 40° C (or 100° F) to 125° C (or 260° F). An International Control and Display Symbol icon for "Engine Coolant Temperature" is located on the cluster overlay, directly above the hub of the gauge needle (Fig. 12). The engine coolant temperature gauge graphics are white and blue against a black field except for a single red graduation at the high end of the gauge scale, making them clearly visible within the instrument cluster in daylight. When illuminated from behind by the panel lamps dimmer controlled cluster illumination lighting with the exterior lamps turned On, the white graphics appear white, the blue graphics appear blue, and the red graphics appear red. The orange gauge needle is internally illuminated. Gauge illumination is provided by replaceable incandescent bulb and bulb holder units located on the instrument cluster electronic circuit board. The engine coolant temperature gauge is serviced as a unit with the instrument cluster.

OPERATION

The engine coolant temperature gauge gives an indication to the vehicle operator of the engine coolant temperature. This gauge is controlled by the instrument cluster circuit board based upon cluster programming and electronic messages received by the cluster from the Powertrain Control Module (PCM) over the Programmable Communications Interface (PCI) data bus. The engine coolant temperature gauge is an air core magnetic unit that receives battery current on the instrument cluster electronic circuit board through the fused ignition switch output (run-start) circuit whenever the ignition switch is in the On or Start positions. The cluster is programmed to move the gauge needle back to the low end of the scale after the ignition switch is turned to the Off position. The instrument cluster circuitry controls the gauge needle position and provides the following features:

- **Engine Temperature Message** - Each time the cluster receives a message from the PCM indicating the engine coolant temperature is between about 40° C (100° F) and 124° C (255° F), the gauge needle is moved to the actual relative temperature position on the gauge scale.
- **Engine Temperature Low Message** - Each time the cluster receives a message from the PCM indicating the engine coolant temperature is at or below about 40° C (100° F), the gauge needle is held at the 40° C (100° F) graduation at the far left end of the gauge scale. The gauge needle remains at the left end of the gauge scale until the cluster receives a message from the PCM indicating that the engine temperature is above about 40° C (100° F), or until the ignition switch is turned to the Off position, whichever occurs first.
- **Engine Temperature High Message** - Each time the cluster receives a message from the PCM indicating the engine coolant temperature is above about 127° C (261° F), the gauge needle is moved into the red zone at the far right end of the gauge scale, the check gauges indicator is illuminated, and a single chime tone is sounded. The gauge needle remains in the red zone and the check gauges indicator remains illuminated until the cluster receives a message from the PCM indicating that the engine temperature is below about 124° C (255° F), or until the ignition switch is turned to the Off position, whichever occurs first. The chime tone feature will occur only once per ignition cycle.
- **Engine Temperature Critical Message** - Each time the cluster receives a message from the PCM indicating the engine coolant temperature is above about 129° C (264° F), the gauge needle is moved to the far right end of the red zone on the gauge scale. The gauge needle remains at the far right end of the red zone until the cluster receives a message from the PCM indicating that the engine temperature is below about 127° C (261° F), or until the ignition switch is turned to the Off position, whichever occurs first.
- **Communication Error** - If the cluster fails to receive an engine temperature message, it will hold the gauge needle at the last indication about twelve seconds or until the ignition switch is turned to the Off position, whichever occurs first. After twelve seconds, the cluster will move the gauge needle to the left end of the gauge scale.
- **Actuator Test** - Each time the cluster is put through the actuator test, the gauge needle will be swept to several calibration points on the gauge scale in a prescribed sequence to confirm the functionality of the gauge and the cluster control circuitry.

The PCM continually monitors the engine coolant temperature sensor to determine the engine operating temperature. The PCM then sends the proper